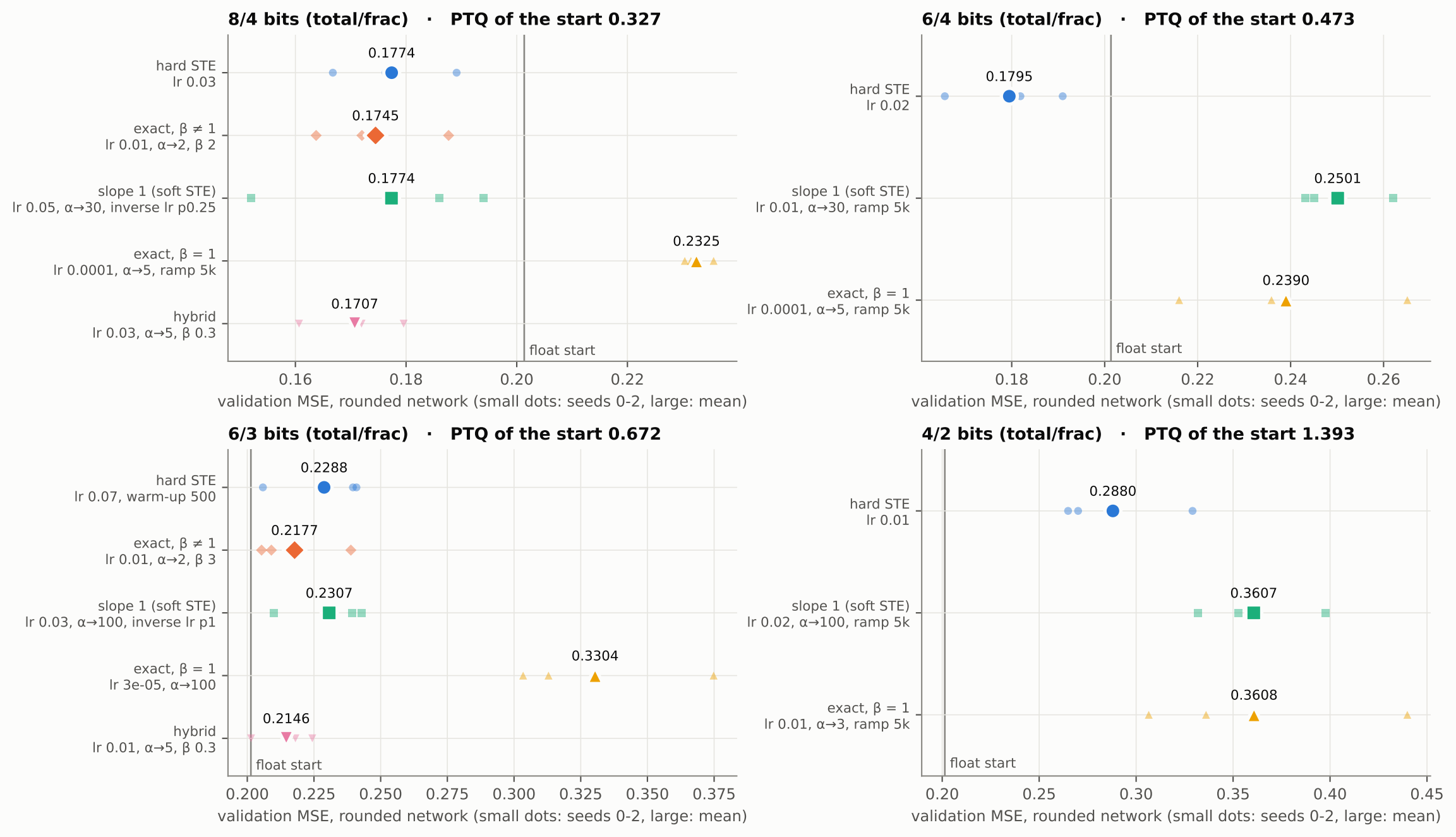
8/4 bits (total/frac) · PTQ of the start 0.327
hard STE
lr 0.03
0.1774
exact, β ≠ 1
lr 0.01, α→2, β 2
0.1745
slope 1 (soft STE)
lr 0.05, α→30, inverse lr p0.25
0.1774
exact, β = 1
lr 0.0001, α→5, ramp 5k
0.2325
hybrid
lr 0.03, α→5, β 0.3
0.1707
float start
0.16
0.18
0.20
0.22
validation MSE, rounded network (small dots: seeds 0-2, large: mean)
6/4 bits (total/frac) · PTQ of the start 0.473
hard STE
lr 0.02
0.1795
slope 1 (soft STE)
lr 0.01, α→30, ramp 5k
0.2501
exact, β = 1
lr 0.0001, α→5, ramp 5k
0.2390
float start
0.18
0.20
0.22
0.24
0.26
validation MSE, rounded network (small dots: seeds 0-2, large: mean)
6/3 bits (total/frac) · PTQ of the start 0.672
hard STE
lr 0.07, warm-up 500
0.2288
exact, β ≠ 1
lr 0.01, α→2, β 3
0.2177
slope 1 (soft STE)
lr 0.03, α→100, inverse lr p1
0.2307
exact, β = 1
lr 3e-05, α→100
0.3304
hybrid
lr 0.01, α→5, β 0.3
0.2146
float start
0.200
0.225
0.250
0.275
0.300
0.325
0.350
0.375
validation MSE, rounded network (small dots: seeds 0-2, large: mean)
4/2 bits (total/frac) · PTQ of the start 1.393
hard STE
lr 0.01
0.2880
slope 1 (soft STE)
lr 0.02, α→100, ramp 5k
0.3607
exact, β = 1
lr 0.01, α→3, ramp 5k
0.3608
float start
0.20
0.25
0.30
0.35
0.40
0.45
validation MSE, rounded network (small dots: seeds 0-2, large: mean)